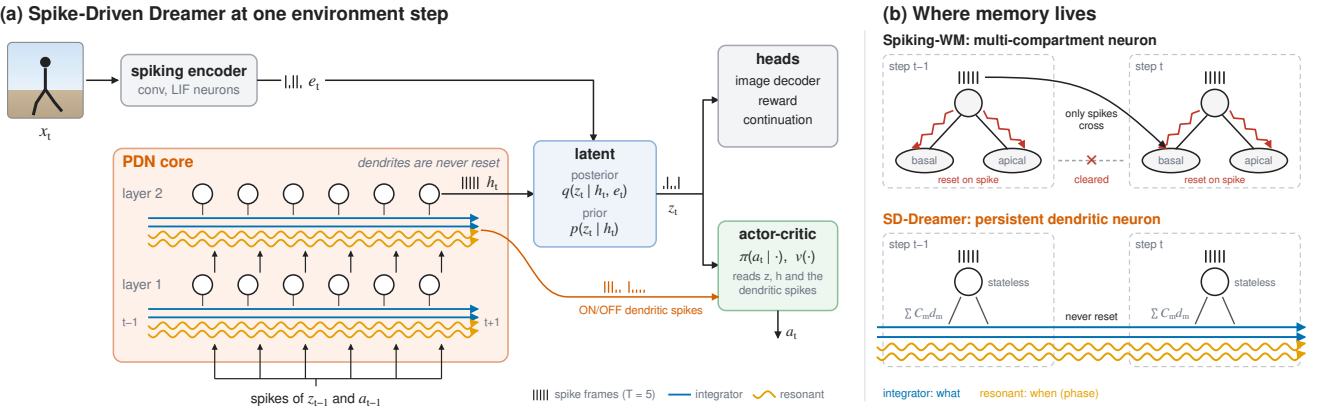

(a) Spike-Driven Dreamer at one environment step
x_t
spiking encoder
conv, LIF neurons
e_t
heads
image decoder
reward
continuation
PDN core
dendrites are never reset
layer 2
layer 1
t−1
t+1
h_t
latent
posterior
$q(z_t \mid h_t, e_t)$
prior
$p(z_t \mid h_t)$
z_t
actor-critic
$\pi(a_t \mid \cdot),\ v(\cdot)$
reads z, h and the dendritic spikes
ON/OFF dendritic spikes
a_t
spikes of z_{t-1} and a_{t-1}
spike frames (T = 5)
integrator
resonant
(b) Where memory lives
Spiking-WM: multi-compartment neuron
step t−1
step t
basal
apical
only spikes cross
reset on spike
cleared
reset on spike
SD-Dreamer: persistent dendritic neuron
stateless
$\sum C_m d_m$
never reset
integrator: what
resonant: when (phase)